Example: *Ecballium elatrium* (Squirting cucumber) *Gyrocarpus* and *Dipterocarpus.*



Figure 6.34: *Ecballium*



Figure 6.35: *Impatents*

Human aided seed dispersal



Figure 6.36: Seed ball

Seed Ball : Seed ball is an ancient Japanese technique of encasing seeds in a mixture of clay and soil humus (also in cow dung) and scattering them on to suitable ground, not planting of trees manually. This method is suitable for barren and degraded lands for tree regeneration and vegetation before monsoon period where the suitable dispersal agents become rare.

Guess? what is atelochory or Achory?

Ecologically important days
March 21 - World forest day
April 22 - Earth day
May 22 - World bio diversity day
June 05 - World environment day
July 07 - Van Mohostav day
September 16 - International Ozone day

Advantages of seed dispersal:

- Seeds escape from mortality near the parent plants due to predation by animals or getting diseases and also avoiding competition.
- Dispersal also gives a chance to occupy favourable sites for growth.
- It is an important process in the movement of plant genes particularly this is the only method available for self-fertilized flowers and maternally transmitted genes in outcrossing plants.
- Seed dispersal by animals help in conservation of many species even in human altered ecosystems.
- Understanding of fruits and seed dispersal acts as a key for proper functioning and establishment of many ecosystems from deserts to evergreen forests and also for the maintenance of biodiversity conservation and restoration of ecosystems.

Summary

Ecology is a division of biology and deals with the study of environment in relation to organisms. Ecology is mainly divided into two branches Autecology and Synecology. The environment (surrounding) includes physical, chemical and biological components. These factors can be classified into living (biotic) and non-living (abiotic), which make the environment of an organism. The ecological factors are meaningfully grouped into four classes, which are as follows: 1. Climatic factors 2. Edaphic factors 3. Topographic factors 4. Biotic factors.

Climate is one of the important natural factors controlling the plant life. The climatic factors includes light, temperature, water, wind, fire, etc. Edaphic factors, the abiotic factors related to soil, include the physical and chemical composition of the soil formed in a particular area. The surface features of earth are called topography. Topographic influence on the climate of any area is determined by the interaction of solar radiation, temperature, humidity ,rainfall, latitude and altitude. The interactions among living organisms, the plants and animals are called biotic factors, which may cause marked effects upon vegetation.

The modifications in the structure of organisms to survive successfully in an environment are called adaptations of organisms. Based on the habitats and the corresponding adaptations of plants, they are classified into 1) Hydrophytes 2) Xerophytes 3) Mesophytes 4) Epiphytes and 5) Halophytes.

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_122-147 CH 06.indd 129 3/14/2020 11:38:47 AM